

Reactive greenhouse gas scenarios: Systematic exploration of uncertainties and the role of atmospheric chemistry

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[1] Knowledge of the atmospheric chemistry of reactive greenhouse gases is needed to accurately quantify the relationship between human activities and climate, and to incorporate uncertainty in our projections of greenhouse gas abundances. We present a method for estimating the fraction of greenhouse gases attributable to human activities, both currently and for future scenarios. Key variables used to calculate the atmospheric chemistry and budgets of major non-CO₂ greenhouse gases are codified along with their uncertainties, and then used to project budgets and abundances under the new climate-change scenarios. This new approach uses our knowledge of changing abundances and lifetimes to estimate current total anthropogenic emissions, independently and possibly more accurately than inventory-based scenarios. We derive a present-day atmospheric lifetime for methane (CH₄) of 9.1 ± 0.9 y and anthropogenic emissions of 352 ± 45 Tg/y (64% of total emissions). For N₂O, corresponding values are 131 ± 10 y and 6.5 ± 1.3 TgN/y (41% of total); and for HFC-134a, the lifetime is 14.2 ± 1.5 y. **Citation:** Prather, M. J., C. D. Holmes, and J. Hsu (2012), Reactive greenhouse gas scenarios: Systematic exploration of uncertainties and the role of atmospheric chemistry, *Geophys. Res. Lett.*, 39, L09803, doi:10.1029/2012GL051440.

[2] Increase in the atmospheric abundances of greenhouse gases (GHG) from pre-industrial era (PI) to present day (PD) is ascribed with high certainty as the major cause of recent climate change [Santer *et al.*, 1993; Hegerl *et al.*, 2007; Barnett *et al.*, 2008; Zwiers and Hegerl, 2008; Min *et al.*, 2011]. Attribution of the observed increases in three dominant long-lived GHG (CO₂, CH₄, N₂O) to human activities, however, has not been made with an equivalent level of certainty. Industrialization and land-use change have unquestionably driven most of these increases, but natural emissions and sinks, including atmospheric chemistry, can also have changed over the industrial era. Formal attribution including uncertainties of the observed GHG changes to humans requires a causal chain relating human activities to emissions to abundances to radiative forcing, one that includes propagation of uncertainties as outlined here.

[3] Projections of the long-lived, chemically controlled greenhouse gases – CH₄, N₂O, and the HFC-134a (CH₂FCF₃) – to year 2100 were made for the Intergovernmental Panel on

Climate Change's (IPCC) Third Assessment Report (TAR) using anthropogenic emissions specified in the Special Report on Emissions Scenarios (SRES) [Nakicenovic *et al.*, 2000]. These included changes in atmospheric chemistry based on a multi-model study of global tropospheric chemistry done for the TAR [Prather *et al.*, 2001]. The IPCC 4th Assessment Report (AR4) [Intergovernmental Panel on Climate Change (IPCC), 2007] kept the SRES scenarios for these GHG and their projections from the TAR. With the IPCC 5th Assessment Report (AR5) being prepared for 2013, the community put together a new set of scenarios, the Representative Concentration Pathways (RCPs) [Moss *et al.*, 2010; van Vuuren *et al.*, 2011], which included a mapping of the emissions onto GHG abundances using a single parametric model [Meinshausen *et al.*, 2011b]. Projected abundances of CO₂ in climate assessments have in general included a measure of uncertainty such as the use of two carbon-cycle models to indicate a range in the TAR Appendix II [Prentice *et al.*, 2001; Reisinger *et al.*, 2010], but for reactive GHG a single best value is assumed. Climate model calculations for the 2013 IPCC AR5 (CMIP5) [Lamarque *et al.*, 2010; Jones *et al.*, 2011; Meinshausen *et al.*, 2011a, 2011b] (CMIP5, Coupled Model Intercomparison Project Phase 5, 2012, available at <http://cmip-pcmdi.llnl.gov/cmip5/>) are using mainly the scientific understanding of atmospheric chemistry and trace-gas budgets from the 2001 TAR to map anthropogenic emissions to abundances for long-lived GHG other than CO₂. Related studies for the AR5, the Atmospheric Chemistry and Climate MIP (ACCMIP), are investigating the relationship between emissions and composition for the reactive gases and aerosols with 3-D chemistry-climate models.

[4] Knowledge of uncertainty in current anthropogenic emissions is critical in projecting future abundances of N₂O and CH₄. For the industrially produced GHG, such as fossil-fuel CO₂ and synthetic HFCs, anthropogenic emissions estimates are generally accurate to 10% or better based on bottom-up inventory methods, although emissions of some synthetic fluorinated gases such as SF₆ remain greatly underestimated [Levin *et al.*, 2010]. Anthropogenic emission estimates from the agriculture, forestry and other land-use sectors (AFOLU), particularly for N₂O and CH₄, have uncertainties typically 25%–50% or larger [Olivier and Berndowski, 2001; van Aardenne *et al.*, 2001; Denman *et al.*, 2007; Prather *et al.*, 2009; National Research Council, 2010]. Thus, inventory-based estimates of these anthropogenic emissions, as in SRES and RCP, have uncertainties that would hinder quantitative attribution of the observed GHG increase to human activities. Both SRES and RCP define a set of scenarios with a range in anthropogenic emissions (e.g., RCP2.6, RCP4.5, RCP6.0, RCP8.5), each based on independent integrated assessment models. In the inventory-based method natural emissions are chosen to be constant and to

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Table 1. Key Data for Pre-industrial and Present-Day Budgets of CH₄ and N₂O^a

Quantity	CH ₄ PI	CH ₄ PD	N ₂ O PI	N ₂ O PD
abundance (ppb)	700 ± 25 (d2)	1795 ± 18 (e2)	270 ± 7 (d1)	323 ± 3.2 (e1)
growth (ppb/y)		5 ± 1 (f2)		0.8 ± 0.1 (f1)
lifetime (y)	9.5 ± 1.3 (P2)	9.1 ± 0.9 (H1)	142 ± 14 (P1)	131 ± 10 (t1)
emissions – total (Tg/y)	202 ± 28 (Q2)	554 ± 56 (K2)	9.1 ± 1.0 (Q1)	15.7 ± 1.1 (K1)
– natural (Tg/y)	202 ± 28 (Q2)	202 ± 35 (R2)	9.1 ± 1.0 (Q1)	9.1 ± 1.3 (R1)
– anthropogenic (Tg/y)		352 ± 45 (S2)		6.5 ± 1.3 (S1)

^aPre-industrial (PI) is year 1750. Present-day (PD) is year 2010. Abundances are tropospheric-mean mole fraction (ppb = 1 part in 10⁹). PI CH₄ abundance does not include bias correction for different PI vs. PD standard scales [Dlugokencky et al., 2005]. Emissions are Tg-CH₄ (10¹² g) for CH₄, and Tg-N for N₂O. Uncertainties are denoted as ± one standard deviation (16%-to-84% = 68% confidence interval, see text). Quantities are selected from the larger list in Tables S1 and S2, with their location in those tables denoted in parentheses. Emissions sums may not balance due to the MC derived values.

exactly fill the gap between total emissions inferred from the model's adopted lifetime and the harmonized anthropogenic emissions, thus ensuring continuity with the observed abundances and trends at the start of the integration [Prather et al., 2001; Meinshausen et al., 2011a].

[5] Our approach to quantifying anthropogenic emissions is quite different from previous assessments and includes uncertainties throughout. Using a best estimate for current lifetime and atmospheric abundance, total PD sources are derived. Using observed pre-industrial abundances and model estimates of the PI-PD change in lifetimes, we calculate PI emissions. Current natural emissions are derived from these plus estimates of the change in natural emissions. For example, one study suggests natural CH₄ emissions may have decreased by 10% due to loss of wetlands [Houweling et al., 2000], and we adopt a ±10% uncertainty in this change. We apportion the present-day budget into natural and anthropogenic sources with correlated uncertainties in each as shown in Table 1.

[6] First, we identify and quantify the key factors controlling past and future abundances of CH₄, N₂O, and HFC-134a. In terms of the Kyoto-specified GHG, other HFCs can be evaluated similarly to HFC-134a, while SF₆ and the perfluorocarbons are so long lived that their 21st century abundances are not impacted by uncertainties in atmospheric lifetimes. Primary factors include observations of past and present atmospheric abundances, recent growth rates, and data required to calculate global burdens and radiative forcing. Table S1 lists primary factors identified by lower-case letters (a)–(v) with numbers (e.g., (d1) = PI abundance of N₂O, (d2) = PI abundance of CH₄).¹ These are generally independent, and we assume that they have uncorrelated uncertainties, denoted in A1 both as ±absolute and ±relative (%) numbers and assumed to be the 16%-to-84% confidence interval and normally distributed (i.e., one-sigma about the mean). The values and uncertainties of the prime factors are derived from published studies, multiple models or lines of evidence, and expert judgment as noted in Table S1. Data derived primarily from atmospheric measurements, including air from polar ice sheets, are given asterisks (c*, d*, e*, f*, k*). Preindustrial (PI) is taken to be year 1750 per IPCC [2007], and present day is year 2010 with current trends based on years 2006–2010.

[7] Derived factors, such as atmospheric burdens, lifetimes and budgets, are listed in Table S2 and designated by upper-

case letters (A)–(S). Their values and uncertainties are calculated from the primary values and their uncertainties. Where terms are additive, an absolute uncertainty is calculated from the square root of the sums of the squares of the absolute uncertainties; where multiplicative, a relative uncertainty is calculated similarly from the relative (%) uncertainties. This quadrature-sum method is included in Table S2 and assumes that all uncertainties are normal or long-normal. As a check we recalculated uncertainties with a Monte Carlo (MC) method: (1) a random number is used to select a single value for a primary factor from the probability distribution defined by the mean and standard deviation; (2) this process is repeated with different random numbers for all 40 primary factors in Table S1; (3) we calculate a single value for each of the 46 derived factors in Table S2; (4) this process is repeated for 100,000 realizations; and (5) the mean and standard deviation in Table S2 are calculated from all realizations. The spreadsheet simple formulae and the MC agree except for emissions (Q–S), where uncertainties in the propagated quantities are highly correlated.

[8] Methylchloroform (MCF: CH₃CCl₃), a synthetic hydrochlorocarbon whose use has been phased out, has become the bellwether of tropospheric chemistry by providing an integrated measure of the reaction with tropospheric hydroxyl radical (OH) [Prinn et al., 1992; Spivakovsky et al., 2000]. The decay of tropospheric MCF since 2006 has followed a simple exponential decay that reflects the total atmospheric lifetime (k1) [Montzka et al., 2011]. The atmospheric lifetime for a process is defined properly as the total atmospheric burden divided by total losses (e.g., against tropospheric OH, stratospheric photolysis), and thus inverse lifetimes (mean loss frequencies) are additive [Forster et al., 2007; Prather, 2007]. By subtracting loss frequencies due to stratospheric photolysis (l1) and oceanic uptake (m1), a mean loss frequency (i.e., inverse lifetime) against tropospheric OH is derived (F1). This lifetime can be scaled to other species that react with OH by their relative reaction rates (h–i) [Prather and Spivakovsky, 1990; Spivakovsky et al., 2000; Lawrence et al., 2001]. Scaling the MCF decay rate to that of CH₄ and HFC-134a by reaction rates (i1–i2) assumes that total atmospheric burden scales with tropospheric mean abundance. The fill factor (c) defined here represents the reduced abundance of a gas in the stratosphere due to loss at high altitudes, it is calculated as the ratio of the integrated burden of the observed profile relative to that if it were uniformly mixed throughout the atmosphere. Thus, we correct the MCF decay with the fill factor (c4) to correspond to the decay for a uniform MCF

¹Auxiliary materials are available in the HTML. doi:10.1029/2012GL051440.

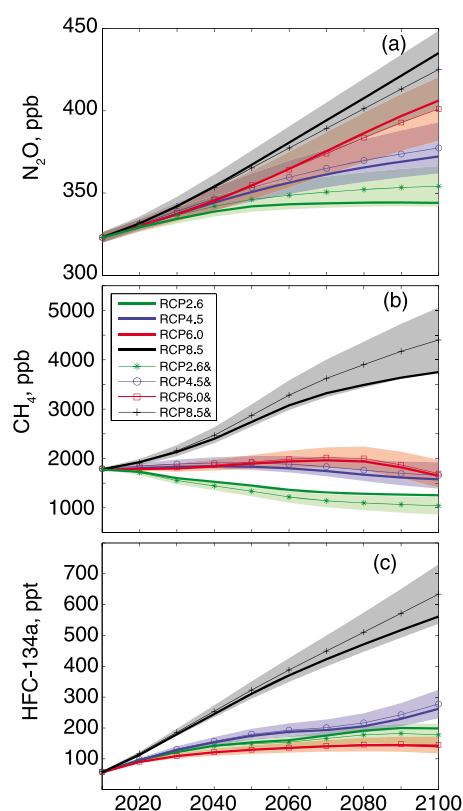


Figure 1. Projected atmospheric abundances for (a) N_2O , (b) CH_4 and (c) HFC-134a based on the four RCP emissions scenarios (2010–2100). The thick solid lines show the published RCP values: black, RCP8.5; red, RCP6.0; blue, RCP4.5; green, RCP 2.6 [Meinshausen *et al.*, 2011b]. Thin lines with markers show mean values derived here with shaded regions showing ± 1 standard deviation based on the Monte Carlo calculations. Our results include all uncertainties listed in Table S1 and rescale the RCP emissions to match our derived year-2010 anthropogenic emissions.

abundance (F2), and then adjust the scaled OH lifetimes of CH_4 and HFC-134a (F3–F5) using their fill factors (c2–c3).

[9] The lifetime of CH_4 derived here, 9.1 ± 0.9 y, is about 5% longer with similar uncertainty as in AR4 [Denman *et al.*, 2007] (8.7 ± 1.3 y, quoted as 90%-confidence interval). Our uncertainty ($\sim 10\%$) is dominated by the ratio of the OH-reaction rates (i1). It would be much larger ($>23\%$) if we took the separate uncertainties for each reaction from JPL-2010 [Sander *et al.*, 2010] as uncorrelated. Instead, we use an average of the recommended rates (i1) but take the uncertainty from the direct measurements of the ratio of these rates by DeMore and co-workers [DeMore, 1992; Hsu and DeMore, 1995], greatly reducing the uncertainty. Large uncertainties in some of the poorly understood, primary data can be readily shown to have little influence. For example, the lifetime of CH_4 against soil uptake (n1) used here ($150 \text{ y} \pm 33\%$) is based on TAR values, but use of a more recent evaluation ($200 \text{ y} \pm 50\%$, [Curry, 2007]) increases the CH_4 lifetime from 9.14 to 9.28 y, well within the one-sigma uncertainty range of $\pm 10\%$.

[10] The total PD emissions for CH_4 and N_2O remain within 5% of those given in AR4 [Denman *et al.*, 2007]. For CH_4 the estimate of PD anthropogenic emissions here,

$352 \pm 45 \text{ Tg/y}$, is larger than that in the RCPs, $\sim 335 \text{ Tg/y}$ (the RCPs, harmonized in year 2005, have a spread of 24 Tg/y by 2010), but well within our uncertainty range. The major sources of uncertainty in our estimate are the PI-PD change in natural emissions and the OH-lifetime, both PI and PD. For N_2O the estimate of PD anthropogenic emissions here, $6.5 \pm 1.3 \text{ Tg-N/y}$, is much smaller than that in the RCPs, $\sim 7.9 \text{ Tg-N/y}$ (the RCPs have a spread of 0.4 Tg-N/y by 2010), which fall just outside the one-sigma range. This difference is due to more recent evaluations of the PD N_2O lifetime [Hsu and Prather, 2010; Fleming *et al.*, 2011]. These top-down constraints on uncertainty in the current total anthropogenic emissions – 13% for CH_4 and 20% for N_2O – are a factor of two better than those derived from bottom-up inventory methods, even for developed countries – 25% for CH_4 and 50% for N_2O [Prather *et al.*, 2009] – but may be optimistic as there is limited research on the uncertainty in PI-PD changes of emissions and lifetime.

[11] With this model we project the future abundances of the greenhouse gases CH_4 , N_2O , and HFC-134a initialized with observed values in year 2010 and integrated to 2100 using the RCP anthropogenic emissions (see Figure 1). For each of the 100,000 MC realizations, a value for PD (year 2010) anthropogenic emissions of CH_4 and N_2O is derived. For each of the four RCPs, all projected emissions of CH_4 and N_2O are scaled by a single factor to match the MC value at 2010, and we assume no future change in natural emissions. To project the change in lifetime to 2100 in this study we use only the self-feedback factors (q1, s1) and very simple estimates with uncertainties of the impacts of other factors on lifetime (r1, u1). A significant, but not assessed, uncertainty in projecting future CH_4 lies with the RCP projections of other anthropogenic reactive gases considered pollutants, such as nitric oxide, carbon monoxide, and aerosols. These species control tropospheric OH, and there remains considerable uncertainty in their future emissions for any scenario [Cofala *et al.*, 2007; Kloster *et al.*, 2010]. For each of the 100,000 MC realizations, a single time-line from 2010 to 2100 is generated for each gas and each RCP. Every decade the MC ensemble mean and spread ($\pm$ standard deviation) are plotted in Figure 1, where our projected CH_4 , N_2O and HFC-134a abundances and uncertainties are compared with the single-valued RCPs [Meinshausen *et al.*, 2011b].

[12] In spite of the 20% difference in anthropogenic N_2O emissions, our abundances track the RCP abundances in Figure 1a for each scenario over the first decades because both have constrained total emissions to match current abundance and growth rate with the chosen lifetime. The differences at 2100 reflect our model's larger self-feedback factor (s1) and longer lifetime (l1): reducing the larger abundances in RCP8.5 and RCP6.0, while increasing the smaller abundances in RCP2.6 and RCP 4.5. Thus, our analysis predicts a tight range of future N_2O abundances for the same range of anthropogenic emissions, but the uncertainty range includes the larger spread of the RCP abundances. Results for CH_4 in Figure 1b show the opposite: both the large increases projected for RCP8.5 and the large decreases projected for RCP2.6 are enhanced in our analysis. The cause is likely similar to that for N_2O , but with differences in lifetime and feedback amplifying the changes, particularly for RCP8.5 with the largest emissions increases. Another possible cause of CH_4 differences in Figure 1b is

that the RCP model includes a temperature dependence of the OH-CH₄ reaction, reducing the lifetime with global warming. This factor was not included here because it would presume that RCPs correctly project climate change and further studies are needed to relate the global mean to the pattern of temperature change over the primary regions where CH₄ is destroyed. As noted above, the synthetic greenhouse gas HFC-134a has no natural emissions and hence the RCP emissions cannot be scaled using this method. The range in projected HFC-134a abundances shown for each RCP in Figure 1c is caused primarily by uncertainty in the tropospheric OH lifetime and its response to changing CH₄ abundance.

[13] Quantitative attribution of a change in N₂O or CH₄ abundances to human activities, from pre-industrial to present day or for future changes, could be made with a statistical model such as this; however, a more careful assessment of uncertainty in the change of natural emissions and lifetime is needed. In addition, uncertainties in inventory-based emissions from the AFOLU sector would need to be included explicitly (e.g., Bayesian methods). Such an approach could make a quantitative attribution to the primary anthropogenic emissions of N₂O and CH₄. The importance of other anthropogenic emissions on the lifetimes (p1, u1) or of global change as a whole (e.g., climate, pollution, land use) should also be included, but will need a wider approach.

[14] The values derived here represent our current best estimates, and we expect them to change with new scientific results and a broader community evaluation of the uncertainties. Even the scope of primary quantities can expand. Thus, we provide in the auxiliary material both the spreadsheet for calculation of many derived quantities and the Matlab code used in the Monte Carlo calculation of emissions. A clear advantage of the approach outlined here is that new science is readily incorporated and that critical variables driving the uncertainty can be identified and become a research focus.

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